

Categorical Embedding for Entity Resolution

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Abstract

Entity Resolution (ER) and Deduplication seek to identify records that refer to the same real-world entity. Contemporary ER systems rely on textual similarity, neural embeddings, or graph-based clustering, yet they rarely model the *categorical structure* inherent in multi-field record systems. We introduce **Categorical Embedding for Entity Resolution (CEER)**, a framework grounded in category theory and motivated by algebraic K -theoretic notions such as homotopy consistency, functoriality, and path invariance. CEER treats records as objects in a category, attribute comparisons as typed morphisms, and learns a functor into a geometric embedding space where consistency across morphism compositions is explicitly encouraged.

The CEER framework consists of three components: (1) a functorial embedding that maps records and attribute morphisms into a vector space, (2) a homotopy-inspired regularization that penalizes inconsistent comparison paths, and (3) a categorical Potts clustering model whose objective incorporates structural constraints derived from the underlying morphisms. Together, these components provide a principled mechanism for reasoning about multi-field similarity, addressing challenges such as field heterogeneity, transitivity violations, and contradictory similarity cycles.

Although empirical evaluation is ongoing, the theoretical formulation and algorithmic design of CEER suggest that categorical structures can serve as a unifying mathematical foundation for ER. CEER is intended to complement rather than replace modern approaches, including large-language-model (LLM) baselines: whereas LLMs reason locally over textual pairs, CEER focuses on global structural consistency and type-aware transformations. This manuscript formalizes the CEER model and outlines its potential advantages for large, noisy, and heterogeneous ER environments.

1 Introduction

Entity Resolution (ER)—the task of determining when multiple records refer to the same real-world entity—remains challenging in heterogeneous, noisy, and large-scale data environments. Modern ER systems span probabilistic linkage models, token- and character-based similarity functions, metric-learning architectures, graph-based clustering, and, more recently, large-language-model (LLM)–assisted matching. Although these approaches differ in representation and computational strategy, they overwhelmingly share a common assumption: a record is a flattened data object whose attributes can be embedded into a unified similarity space, and clustering can be carried out on the resulting pairwise similarities.

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This geometric paradigm, broadly effective in practice, carries structural limitations. First, existing metric-learning models (e.g., contrastive learning, triplet losses) and graph neural network (GNN)-based ER pipelines encode relationships between records but do not distinguish the semantic roles of different fields. A **Title** morphism in a bibliographic record and a **BirthYear** discrepancy in a voter file induce fundamentally different comparison behaviors, yet most ER architectures use monolithic embeddings or homogeneous message passing. Second, these systems do not articulate how multi-step comparison chains should behave relatively to direct comparisons. For example:

Record A and B match strongly on **Title**; B and C match strongly on **Authors**; A and C disagree sharply on **Year**.

Metric-learning models may encode these signals, but they do not specify what consistency relation should hold among the three pairwise embeddings. Graph clustering algorithms (e.g., CPM, Leiden, Louvain) treat such edges as independent and may produce unstable partitions when local decisions conflict.

These observations motivate the present work: a framework in which *field heterogeneity*, *comparison composition*, and *global consistency* receive explicit mathematical treatment.

A constrained categorical perspective. We introduce **Categorical Embedding for Entity Resolution (CEER)**, a framework that treats records as objects and field-specific comparisons as *typed morphisms*. Unlike prior categorical or diagrammatic approaches in machine learning, the goal here is not to establish a full categorical semantics for ER, but to impose a minimal, computationally tractable structure that captures three requirements:

1. **Typed transformations:** comparisons on different fields induce operators of different functional classes (e.g., linear, diagonal, shallow nonlinear).
2. **Approximate composition:** when two fields induce a comparison chain, the corresponding operators should behave coherently with a direct comparison on a third field.
3. **Structural regularity:** embeddings should avoid contradictory similarity cycles that arise when noisy attributes distort geometric neighborhoods.

The categorical terminology serves as a scaffolding for these constraints: each field yields a family of morphisms; operator composition encodes how comparison signals propagate; identities represent neutral comparisons. We do not assume strict functoriality or categorical equivalence; instead, we impose *approximate* functorial behavior through regularization terms that quantify how far a multi-step transformation deviates from a direct transformation.

From homotopy-inspired consistency to practical ER. Homotopy ideas motivate a tolerance-based formulation of consistency: two comparison paths between the same pair of records should produce embeddings that differ by a small deformation rather than an arbitrary discrepancy. This perspective does not claim that ER data obeys topological equivalence; rather, it provides a principled mechanism to penalize similarity triangles that are internally contradictory. This is especially useful in domains such as voter registration files and bibliographic datasets, where noise in a single field can distort entire clusters.

The CEER framework. The complete CEER system consists of three interacting components:

1. **A functorial embedding module** that maps each record to a vector and each typed morphism to an operator, enforcing approximate composition through regularization.
2. **A homotopy-inspired consistency loss** that identifies and penalizes comparison paths whose multi-step transformations deviate significantly from direct transformations.
3. **A categorical Potts clustering objective** that integrates typed morphism information directly into the clustering energy, improving global coherence relative to purely geometric or graph-local methods.

Position in the ER landscape. CEER is not a replacement for metric-learning models, GNN-based ER, or LLM-driven pipelines. Instead, it provides a structural mechanism absent from these systems: a way to encode how different fields interact and how multi-step similarities constrain one another. LLMs excel at local pairwise reasoning, but they do not enforce consistency across millions of comparisons; CEER fills precisely this gap by imposing global regularity constraints.

Contributions. This manuscript makes the following contributions:

- A minimal but mathematically coherent categorical formulation of typed field comparisons for ER.
- A functorial embedding objective that captures approximate operator composition across heterogeneous fields.
- A homotopy-inspired regularization principle for resolving contradictory similarity cycles.
- A categorical Potts model that integrates operator types into clustering, improving structural coherence.

The remainder of the manuscript develops these components in detail, contrasts CEER with existing ER paradigms, and presents empirical evidence across four datasets.

2 Background and Motivation

Entity Resolution (ER) encompasses a broad landscape of models that compare, score, and cluster records originating from structured, semi-structured, or unstructured datasets. Most ER pipelines treat attributes as components of a unified feature space, whether constructed via tokenization, TF-IDF vectors, neural embeddings, or composite similarity measures. While effective, this paradigm obscures differences in the semantic roles that individual fields play in determining equivalence. This section establishes the conceptual motivation for CEER by examining how standard ER systems handle field heterogeneity, how multi-step comparison inconsistencies arise, and how categorical and homotopy-inspired ideas offer a principled mechanism for constraining these inconsistencies.

2.1 Categorical Structures in ER

Classical ER algorithms compute attribute-wise similarities and aggregate them into a single score, implicitly assuming that all fields contribute to the same representational geometry. More recent approaches—including contrastive metric learning, triplet-loss systems, graph-based clustering, and GNN architectures—operate on richer spaces but still treat field comparisons as homogeneous operators. In these systems, the **Title** field of a bibliographic record and the **BirthYear** field of a voter record influence the final representation in the same functional manner: both are converted into vectors, combined, and used for geometric decisions. The internal semantics of fields are not preserved.

To motivate CEER’s alternative perspective, consider three CORA citation records:

- A: same **Title** as B; mismatching **Year** with C.
- B: same **Title** as A; same **Authors** as C.
- C: shares **Authors** with B but differs in **Year** from A.

Standard pipelines produce a triangle of similarities whose clustering outcome depends on thresholds, initialization, or graph-local modularity heuristics. None of these methods articulate how the chain “match on title, then match on authors” should relate to the direct, conflicting comparison “mismatch on year.”

A minimal categorical viewpoint offers a corrective structure. Let C_{rec} denote a category whose objects are records and whose morphisms correspond to *typed* attribute-level comparisons. Each field α induces a family of morphisms $f_\alpha : r_i \rightarrow r_j$ associated with field-specific transformations or similarity computations. The crucial point is not to impose full categorical semantics, but to capture two functional requirements:

1. **Typed comparisons:** the map induced by field α is distinct from that for field β .
2. **Approximate composition:** the effect of performing f_α followed by f_β should relate predictably to the effect of a direct comparison on any field whose semantics connect r_i and r_k .

This limited categorical structure addresses a central limitation of standard ER systems: the absence of a principled mechanism to express how comparisons along different fields interact.

2.2 Homotopy-Inspired Consistency and Multi-Step Comparisons

Real ER datasets frequently exhibit structural inconsistencies. In a voter registration file, two records may match on name and address, while a third matches on name but contradicts both on birth year. Classical and neural ER pipelines detect high pairwise similarity along specific edges, but they do not articulate how multi-step comparison paths constrain or contradict direct comparisons.

To illustrate, consider four records in a noisy administrative file:

- r_i : “SAM CHU”, address “123 OAK ST”, birth year 1985,
- r_j : normalized variant of r_i ,
- r_k : same name, partially normalized address, different county,
- r_ℓ : same name as r_k , but different birth year.

Edges $r_i \rightarrow r_j$ and $r_j \rightarrow r_k$ may be strong, but if $r_i \rightarrow r_k$ is weak due to the county change, the resulting triangle is contradictory. Graph-based community detection (e.g., CPM, Leiden) may still cluster all three nodes, or may exclude r_k , depending on initialization and scoring nuances. The key issue is the absence of constraints that enforce coherence among paths.

CEER introduces a homotopy-inspired regularization principle to quantify and penalize these contradictions. Rather than invoking strict topological homotopy, we define an *approximate deformation tolerance*: for two comparison paths between the same records, the corresponding transformations in embedding space should differ by only a small, structured deviation. Let T_α and T_β denote operators associated with fields α and β . Then, for a two-step path $r_i \xrightarrow{f_\alpha} r_j \xrightarrow{f_\beta} r_k$ and a direct comparison $r_i \xrightarrow{f_\gamma} r_k$, CEER imposes:

$$\|T_\beta(T_\alpha(z_i)) - T_\gamma(z_i)\|^2 \quad \text{small.}$$

This formulation does not assume that data obeys algebraic topology; it simply provides a principled quantitative measure of path inconsistency. In practice, these penalties reduce the prevalence of “broken triangles” that often destabilize clustering outcomes.

2.3 Typed Morphisms as a Minimal Structural Breakthrough

The preceding considerations lead to CEER’s central modeling innovation: *records are objects, and field-specific comparisons are typed morphisms with operator-level representations*. This representation retains three desirable properties:

1. **Field semantics are preserved:** title, name, address, and numeric codes correspond to distinct operator families.
2. **Comparison chains acquire structure:** multi-step comparisons cannot diverge arbitrarily from direct comparisons.
3. **Embedding behaviors become constrained:** operator composition shapes geometric proximity, mitigating the influence of noisy or contradictory fields.

This minimalist categorical structure does not claim expressive completeness. It instead offers a computationally feasible mechanism for encoding the interaction of fields, imposing weak constraints inspired by functoriality and deformation invariance. By combining typed morphisms with operator families and consistency regularization, CEER bridges a gap between purely geometric ER and global structural reasoning, enabling more stable and interpretable equivalence classes in heterogeneous datasets.

3 The Categorical Embedding for Entity Resolution (CEER) Model

Building on the limited categorical structure and homotopy-inspired consistency developed in Section 2, this section formalizes the CEER model. The goal is not to construct a full categorical semantics for ER, but to design a computational framework in which typed morphisms, comparison composition, and consistency constraints can be expressed explicitly and optimized efficiently. The CEER architecture consists of three components:

1. a record category equipped with typed morphisms;

2. a functorial embedding module that maps records and morphisms to operators;
3. a clustering stage based on a categorical Potts objective.

Each component is specified below with minimal assumptions to ensure mathematical coherence and computational realizability.

3.1 Record Category and Typed Morphisms

Let C_{rec} be a small category whose objects are records $\{r_i\}$ drawn from a dataset with schema $\mathcal{A}$ (e.g., **Title**, **Authors**, **Address**, **Year**). For each field $\alpha \in \mathcal{A}$ we introduce a family of morphisms

$$\text{Hom}_\alpha(r_i, r_j) = \{f_\alpha^{(k)} : r_i \rightarrow r_j \mid k \in K_\alpha\},$$

where k indexes field-specific transformations such as normalization operations or similarity modes (token overlap, edit distance, neural similarity, etc.).

In practice, CEER simplifies this potentially large set of morphisms by associating a *single operator family* to each field type. For each pair of records (r_i, r_j) and field α , we compute:

1. a field-specific similarity score $s_\alpha(r_i, r_j) \in [0, 1]$,
2. a morphism-induced operator T_α acting on embeddings.

The categorical structure is intentionally minimal: identities exist but carry no computational effect; composition is defined symbolically by $f_\beta \circ f_\alpha$, and CEER enforces *approximate* coherence between composed and direct comparisons.

3.2 Functorial Embedding Module

CEER defines a mapping

$$F : C_{\text{rec}} \rightarrow \text{Vec}_{\mathbb{R}}^d,$$

where each object r_i is assigned an embedding $z_i \in \mathbb{R}^d$ and each morphism f_α is assigned an operator T_α acting on this space.

Record encoder. Each record r_i is first encoded into a base representation x_i using TF-IDF, lightweight neural embeddings, or hybrid models. A learned projection $W \in \mathbb{R}^{d \times m}$ maps $x_i \mapsto z_i$.

Operator families. To balance expressiveness and identifiability, CEER restricts each field type to one of the following operator classes:

- **Linear operators:** $T_\alpha(z) = W_\alpha z$, for short text fields such as **Title**, **Name**, or **Authors**.
- **Shallow nonlinear operators:**

$$T_\alpha(z) = W_{\alpha,2} \sigma(W_{\alpha,1} z + b_{\alpha,1}) + b_{\alpha,2},$$

used for structured address fields.

- **Diagonal operators:** $T_\alpha(z) = D_\alpha z$, for numeric or categorical fields where attribute mismatches should induce controlled warping (e.g., **Year**, **Zipcode**, **County**).

These operator classes ensure that typed morphisms remain distinguishable while avoiding the unconstrained variability of deep networks.

Approximate functoriality. Functoriality in a strict categorical sense requires

$$F(f_\beta \circ f_\alpha) = F(f_\beta) \circ F(f_\alpha),$$

which is unrealistic for noisy ER datasets. Instead, CEER imposes a *composition deviation* penalty:

$$L_{\text{functor}} = \sum_{\alpha, \beta} \mathbb{E}_i \|T_\beta(T_\alpha(z_i)) - T_{\beta \circ \alpha}(z_i)\|^2.$$

The operators $T_{\beta \circ \alpha}$ are learned parameters representing a direct comparison consistent with the two-step chain (α, β) . This penalty constrains the geometry of multi-step comparisons without forcing exact categorical equations.

3.3 Homotopy-Inspired Path Consistency

Beyond pairwise consistency, CEER enforces coherence among comparison *triangles*. Given a triplet (r_i, r_j, r_k) with field types (α, β, γ) responsible for pairwise similarities, we penalize inconsistent triangles via

$$L_{\text{hom}} = \sum_{i, j, k} \|T_\beta(T_\alpha(z_i)) - T_\gamma(z_j)\|^2.$$

Unlike topological homotopy, which classifies paths up to deformation, CEER uses this construction as a stability constraint: multi-step transformations should not diverge significantly from direct transformations. This regularization reduces contradictory similarity cycles that destabilize clustering under geometric or graph-based ER methods.

3.4 Similarity Graph Construction

After learning embeddings $\{z_i\}$ and operators $\{T_\alpha\}$, CEER constructs a similarity graph with weights

$$S_{ij} = \sigma \left(-\|z_i - z_j\|^2 + \sum_{\alpha} w_{\alpha} s_{\alpha}(r_i, r_j) + \sum_{\alpha} \langle T_{\alpha}(z_i), z_j \rangle \right),$$

where σ is a logistic activation and w_{α} are learned or tuned importance weights. This formulation integrates:

- geometric proximity,
- typed field similarities,
- operator-induced compatibility scores,
- implicit consistency constraints from L_{functor} and L_{hom} .

The resulting graphs are empirically less prone to broken triangles and threshold sensitivity than those constructed solely from TF-IDF or SBERT distances.

3.5 Categorical Potts Clustering

To convert similarity graphs into ER clusters, CEER employs a categorical extension of the Constant Potts Model (CPM). Let c_i denote the cluster label for record i . The energy function is

$$E(C) = - \sum_{i < j} S_{ij} \mathbf{1}(c_i = c_j) + \gamma |C| + \eta \sum_{(i,j,k) \in \mathcal{T}} \Delta_{ijk},$$

where γ controls cluster granularity, η regulates triangle penalties, and $\mathcal{T}$ samples triangles incident on comparison inconsistencies.

For each triangle,

$$\Delta_{ijk} = \max\{0, S_{ij} + S_{jk} - S_{ik} - \tau_{\text{tri}}\},$$

where τ_{tri} sets the tolerance for allowable inconsistencies.

Optimization follows a modified Leiden procedure that incorporates triangle penalties into local gain computations and refinement phases.

3.6 Theoretical and Practical Considerations

CEER operates under minimal assumptions:

- typed operators enforce field-specific transformations;
- composition penalties constrain multi-step comparison behavior;
- homotopy-inspired penalties stabilize triangle structures;
- clustering integrates these structural components globally.

This architecture provides a structured alternative to purely geometric ER while remaining computationally tractable for large administrative datasets.

4 Datasets

We evaluate CEER on four datasets that span structured, semi-structured, synthetic, and large-scale administrative regimes. These datasets differ not only in schema complexity but also in the nature of their ambiguities, noise distributions, and field heterogeneity. This diversity allows us to assess the behavior of typed morphisms, operator families, and path-consistency regularization under substantially different ER conditions.

4.1 Affiliation Dataset

The Affiliation dataset contains 2,260 structured records representing institutional and geographic entities. Each record contains the following canonical fields:

- Name
- Street
- City
- State

- Zipcode
- Country

Ambiguities arise from inconsistent abbreviations, heterogeneous formatting, Unicode noise, and hierarchical geographic components. The dataset contains numerous near-duplicates that differ by small variations in address subfields (e.g., “St” vs. “Street”, differing ZIP+4 fragments).

From CEER’s perspective, Affiliation highlights the interaction of *structured address fields*, where operator classes differ by semantic type. Linear operators suit short text fields such as **Name**, whereas shallow nonlinear operators (two-layer MLPs) capture the combinatorial variation in multilevel address fields. Diagonal operators for **Zipcode** and **State** enforce controlled deformation of the embedding, preventing numeric or code-like attributes from dominating similarity. The dataset’s clean schema and moderate size make it a controlled environment for evaluating typed morphisms and functorial constraints.

4.2 Cora Bibliographic Dataset

The Cora citation dataset contains 1,879 records with up to 16 attributes, including:

- Title, Authors, Address
- Journal, Publisher, Year
- Volume, Pages

Cora is characterized by subtle inconsistencies in author formatting, partial or corrupted meta-data, missing fields, and high redundancy across sources. These ambiguities generate contradictory similarity triangles: two records may share a title and authorship patterns, yet mismatch on publication year or publisher. Such inconsistencies represent precisely the type of multi-step comparison distortions that CEER’s approximate composition and homotopy-inspired penalties aim to stabilize.

Typed morphisms play a crucial role here:

- linear operators for **Title, Authors, Journal, Publisher**;
- shallow nonlinear operators for **Address**;
- diagonal operators for numeric fields such as **Year, Volume, Pages**.

These operator families prevent strongly correlated textual fields from dominating numerically meaningful discrepancies (e.g., publication year). Cora therefore serves as a benchmark for CEER’s capacity to maintain structural coherence across heterogeneous attributes.

4.3 D10K Synthetic Dataset

The D10K dataset contains 10,000 synthetically generated records, each consisting of a single composite text field subject to controlled corruption mechanisms:

- token shuffling,
- character deletions and insertions,
- synonym replacement,
- structured perturbations simulating partial field corruption.

Because D10K lacks multi-field structure, it functions as a *stress test* for CEER’s ability to extract latent equivalence classes when typed morphisms collapse to a single operator family. Here CEER reduces to learning one operator T_{text} and enforcing consistent deformation across synthetically perturbed text. This regime isolates the embedding behavior from the categorical structure and tests whether homotopy-inspired penalties can mitigate contradictory signals under adversarial noise. D10K clarifies the boundary between CEER’s typed-morphism design and its behavior in minimal schematic contexts.

4.4 North Carolina Voter Registration (NCVR) Dataset

The NCVR dataset comprises millions of administrative voter registration records. A typical snapshot includes 7–8 million entries, each with demographic and address-related attributes, such as:

- `FirstName`, `MiddleName`, `LastName`
- `ResidentialAddress`, `MailingAddress`
- `County`, `VoterStatus`, `BirthYear`

NCVR presents extreme heterogeneity and noise:

- address abbreviations and partially missing fields,
- name variations and inconsistent concatenation,
- county mismatches due to relocation or registration errors,
- multiple historical registrations referring to the same individual.

Because full-scale pairwise comparison is infeasible at millions of records, blocking strategies (e.g., county+ZIP) reduce computational complexity. Within each block, CEER leverages typed morphisms sensitive to field semantics:

- linear operators for personal name fields,
- nonlinear operators for `ResidentialAddress` and `MailingAddress`,
- diagonal operators for `County`, `VoterStatus`, and `BirthYear`.

NCVR is a demanding environment for multi-step comparison stability. Contradictory similarity chains arise frequently (e.g., matching on names and partial addresses but mismatching on birth year), and CEER’s homotopy-inspired penalties counteract the accumulation of such distortions. This dataset tests scalability, stability under severe noise, and the interaction of homotopy penalties with categorical Potts clustering.

4.5 Summary

These four datasets represent complementary ER challenges:

- structured multi-field consistency (Affiliation),
- heterogeneous bibliographic metadata with multi-step inconsistencies (Cora),

- unstructured adversarial noise (D10K),
- large-scale administrative heterogeneity and blocking requirements (NCVR).

Together, they evaluate whether typed morphisms, operator families, and approximate path-consistency constraints improve ER performance across diverse contexts.

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